

$$1) P = 50 \text{ kN} \quad \text{ASTM A490}$$

Case 2:

$$F_v = 50 \left(\frac{4}{5} \right) = 40 \text{ kN}$$

$$F_H = 50 \left(\frac{3}{5} \right) = 30 \text{ kN}$$

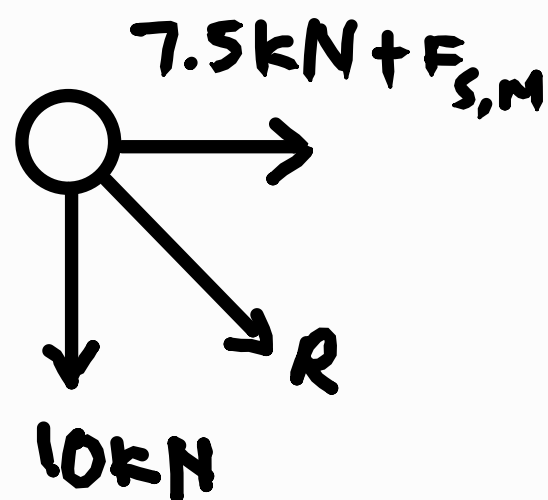
$$F_{s,v} = \frac{F_v}{n} = \frac{40}{4} = 10 \text{ kN}$$

$$F_{s,h} = \frac{F_s}{n} = \frac{30}{4} = 7.5 \text{ kN}$$

$$\begin{aligned} M_c &= F_v d_v + F_H d_H \\ &= 40 \times 250 - 30 \times 50 \\ &= 8500 \text{ Nm} \end{aligned}$$

$$\begin{aligned} F_{s,m} &= \frac{M r_1}{r_1^2 + r_2^2 + r_3^2 + r_4^2} \\ &= \frac{M r_1}{2r_1^2 + 2r_2^2} \quad \because r_1 = r_4 \\ &\quad r_2 = r_3 \\ &= \frac{8500(0.15)}{2(0.15)^2 + 2(0.05)^2} \\ &= 25500 \text{ N} \\ &= 25.5 \text{ kN} \end{aligned}$$

Bolt 1:



Bolt 1 experiences the highest shear force due to being the furthest from C.

$$1) F_1 = \sqrt{10^2 + (7.5 + 25.5)^2}$$

$$= 34.48 \text{ kN}$$

From table 20-1, $\tau_{\max} = 152 \text{ MPa}$

$$\tau_{\max} = \frac{F_T}{A_s}$$

$$\tau_{\max} = \frac{4F_T}{\pi d_r^2}$$

$$d_r = \left(\frac{4F_T}{\pi \tau_{\max}} \right)^{\frac{1}{2}}$$

$$= \left(\frac{4(34.48 \times 10^3)}{152 \times 10^6 \pi} \right)^{\frac{1}{2}}$$

$$= 0.017 = 17 \text{ mm}$$

$$d = \frac{d_r}{0.8} = \frac{17}{0.8} = 21.25 \text{ mm}$$

From table 18-5, M24 bolts

are selected as $d_{\text{selected}} > d_{\text{calculated}}$

$$2) P = 25 \text{ kN} \quad n = 4 \quad \sigma_{\max} = 75 \text{ MPa}$$

Case 1:

$$P_{td} = \frac{P}{n} = \frac{25}{4} = 6.25 \text{ kN}$$

Bolt B experiences the highest force due to it being the furthest away from O.

$$\begin{aligned} P_{tm, B} &= \frac{PL(L_B)}{2L_A^2 + 2L_B^2} \\ &= \frac{(25 \times 10^3)(400 \times 10^{-3})(200 \times 10^{-3})}{2(50 \times 10^{-3})^2 + 2(200 \times 10^{-3})^2} \\ &= 23.53 \times 10^3 \text{ N} = 23.53 \text{ kN} \end{aligned}$$

$$\begin{aligned} P_B &= P_{td} + P_{tm, B} \\ &= 6.25 + 23.53 \\ &= 29.78 \text{ kN} \end{aligned}$$

$$\sigma_{\max} = \frac{P_B}{A}$$

$$\sigma_{\max} = \frac{P_B}{\frac{\pi d_r^2}{4}}$$

$$\begin{aligned} d_r &= \left(\frac{4P_B}{\pi \sigma_{\max}} \right)^{\frac{1}{2}} = \left(\frac{4(29.78 \times 10^3)}{\pi (75 \times 10^6)} \right)^{\frac{1}{2}} \\ &= 0.2249 = 22.49 \text{ mm} \end{aligned}$$

$$2) \quad d = \frac{d_r}{0.8} = \frac{22.49}{0.8} = 28.10 \text{ mm}$$

From table 18-5, M30 bolts are selected.